

Lab Report No 4
Arithmetic Operations
Microprocessor and Interfacing techniques



Submitted By:

Huzaifa Shahbaz

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Submitted to:

Engr. Bushra Fiaz

Dated:

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Department of Computer Engineering,
HITEC University, Taxila

Lab Task No 1(i):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Multiplication)

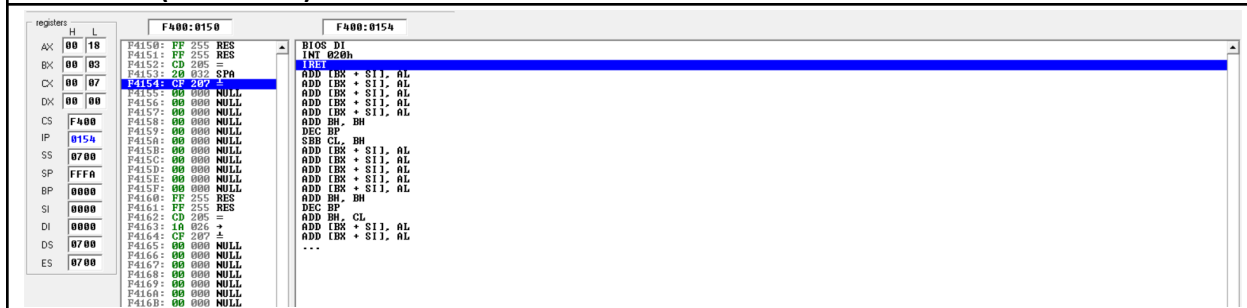
Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes

8*3

The code

```
01 org 100h
02 MOV AL, 8      ; Load the first operand into AL
03 MOV BL, 3      ; Load the second operand into BL
04 MUL BL
05 ret |
```

The results (Screenshot)



Lab Task No 1(ii):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Multiplication)

Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes

2.256*768

The code

```

01 org 100h
02 MOU AX, 2256
03 MOU BX, 768
04 MUL BX
05 ret

```

The results (Screenshot)

Lab Task No 1(iii):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Multiplication)
 Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes
 3. -5*4

The code

```

01 org 100h
02 MOU AX, 3
03 MOU BX, -5
04 MUL BX
05 MOU BX, 4
06 MUL BX
07 ret

```

The results (Screenshot)

Lab Task No 1(iv):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Multiplication)

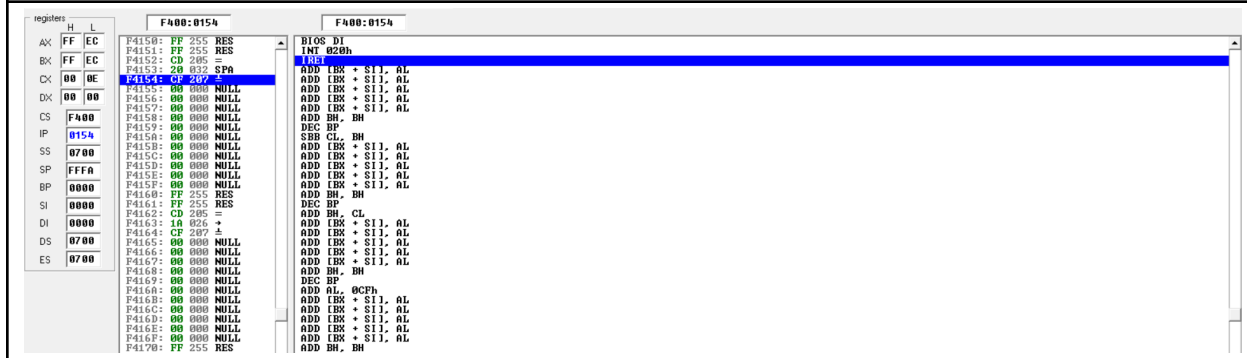
Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes

4.-10*-20

The code

```
01 org 100h
02 MOV AX, 4
03 MOV BX, -10
04 MOV AX, BX
05 MOV BX, -20
06 MOV AX, BX
07 ret
```

The results (Screenshot)



Lab Task No 1(v):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Multiplication)

Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes

5.-256*-768

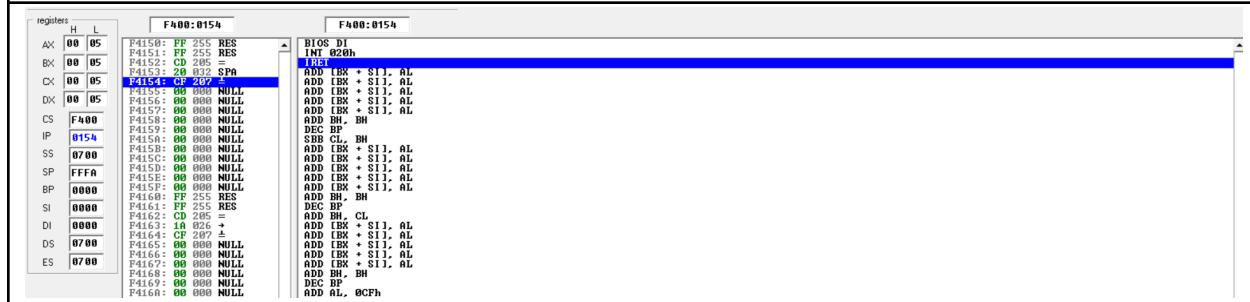
The code

```

01 org 100h
02 MOV AX, 5
03 MOV BX, -256
04 MOV BX
05 MOV CX, AX
06 MOV BX, -768
07 MOV BX
08 MOV DX, AX
09 ret

```

The results (Screenshot)



Lab Task No 2(i):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Division)

Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes, find the remainder and quotient for each expression

1. 10/2

The code

```

01 org 100h
02 MOV AH, DL
03 MOV DL, AH
04 ADD DL, '0'
05 INT 21h
06 MOV DL, DH
07 ADD DL, '0'
08 INT 21h
09 ret

```

The results (Screenshot)

registers		F400:0200		F400:0204	
H	L				
AX	00 00	F4200: FF 255	RES	BIOS DI	
BX	00 00	F4201: FF 255	RES	INT 021h	
CX	00 11	F4202: CD 205	=		
DX	00 30	F4203: 21 033	↑	ADD [BX + SI], AL	
SI		F4204: 00 000	NULL	ADD [BX + SI], AL	
CS	F400	F4205: 00 000	NULL	ADD [BX + SI], AL	
IP	0204	F4206: 00 000	NULL	ADD [BX + SI], AL	
SS	0700	F4207: 00 000	NULL	ADD [BX + SI], AL	
SP	FFFF	F4208: 00 000	NULL	ADD [BX + SI], AL	
BP	0000	F4209: 00 000	NULL	ADD [BX + SI], AL	
SI	0000	F420A: 00 000	NULL	ADD [BX + SI], AL	
DI	0000	F420B: 00 000	NULL	ADD [BX + SI], AL	
DS	0700	F420C: 00 000	NULL	ADD [BX + SI], AL	
ES	0700	F420D: 00 000	NULL	ADD [BX + SI], AL	

Lab Task No 2(ii):

Solution:

Brief description (3-5 lines)

Arithmetic Operations (Division)

Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes, find the remainder and quotient for each expression

2. 15/9

The code

```

01 org 100h
02 MOV AX, 15
03 MOV BX, 9
04 DIV BX
05 MOV AH, 2
06 MOV DL, AH
07 ADD DL, '0'
08 INT 21h
09 MOV DL, DH
10 ADD DL, '0'
11 INT 21h
12 ret

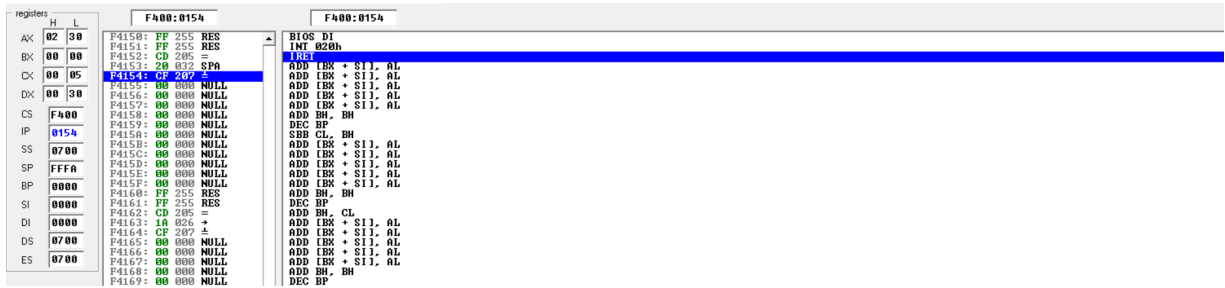
```

The results (Screenshot)

registers		F400:0154		F400:0154	
H	L				
AX	02 30	F4150: FF 255	RES	BIOS DI	
BX	00 09	F4151: FF 255	RES	INT 020h	
CX	00 19	F4152: CD 205	=		
DX	00 30	F4153: 20 032	SPa		
SI		F4154: 00 000	NULL	ADD [BX + SI], AL	
CS	F400	F4155: 00 000	NULL	ADD [BX + SI], AL	
IP	0154	F4156: 00 000	NULL	ADD [BX + SI], AL	
SS	0700	F4157: 00 000	NULL	ADD [BX + SI], AL	
SP	FFFF	F4158: 00 000	NULL	ADD [BX + SI], AL	
BP	0000	F4159: 00 000	NULL	ADD [BX + SI], AL	
SI	0000	F415A: 00 000	NULL	ADD [BX + SI], AL	
DI	0000	F415B: 00 000	NULL	ADD [BX + SI], AL	
DS	0700	F415C: 00 000	NULL	ADD [BX + SI], AL	
ES	0700	F415D: 00 000	NULL	ADD [BX + SI], AL	

Lab Task No 2(iii):

Solution:

Brief description (3-5 lines)		
Arithmetic Operations (Division) Perform following Arithmetic Operations using both Accumulator by Register and Accumulator by Memory modes, find the remainder and quotient for each expression 3. 3844/5		
The code		
<pre>01 org 100h 02 MOV AX, 3844 03 MOV CX, 5 04 DIV CX 05 MOV AH, 2 06 MOV DL, AH 07 ADD DL, '0' 08 INT 21h 09 MOV DL, DH 10 ADD DL, '0' 11 INT 21h 12 ret </pre>		
The results (Screenshot)		
		

Lab Task No 3(i):

Solution:

Brief description (3-5 lines)		
Arithmetic Expressions An arithmetic expression is a mathematical statement that combines numbers and operators, like + (addition), - (subtraction), * (multiplication), or / (division), to perform calculations. 1.CX=(15*10)+20		
The code		

```

01 org 100h
02 MOV AX, 15
03 MOV 10
04 ADD AX, 20
05 MOV CX, AX
06 MOV AH, 2
07 MOV DL, CH
08 ADD DL, '0'
09 INT 21h
10 MOV DL, CL
11 ADD DL, '0'
12 INT 21h
13 ret

```

The results (Screenshot)

registers		F400:0154		F400:0154	
	H	L			
AX	02	4E	F4150: FF 255 RES	BIOS DI	
BX	00	00	F4151: FF 255 RES	INT 020h	
CX	00	1E	F4152: CD 205 =	INT	
DX	00	4E	F4153: 20 032 SP0	ADD EBX + SI, AL	
CS	F400		F4154: CF 207 ±	ADD EBX + SI, AL	
IP	0154		F4155: 00 000 NULL	ADD EBX + SI, AL	
SS	0700		F4156: 00 000 NULL	ADD EBX + SI, AL	
SP	FFFA		F4157: 00 000 NULL	ADD EBX + SI, AL	
BP	0000		F4158: 00 000 NULL	ADD BH, BH	
SI	0000		F4159: 00 000 NULL	DEC BP	
DI	0000		F415A: 00 000 NULL	SBB CL, BH	
DS	0700		F415B: 00 000 NULL	ADD EBX + SI, AL	
ES	0700		F415C: 00 000 NULL	ADD EBX + SI, AL	
			F415D: 00 000 NULL	ADD EBX + SI, AL	
			F415E: 00 000 NULL	ADD EBX + SI, AL	
			F415F: 00 000 NULL	ADD EBX + SI, AL	
			F4160: FF 255 RES	ADD BH, BH	
			F4161: FF 255 RES	DEC BP	
			F4162: CD 205 =	ADD BH, CL	
			F4163: 1A 026 +	ADD EBX + SI, AL	
			F4164: CF 207 ±	ADD EBX + SI, AL	
			F4165: 00 000 NULL	ADD EBX + SI, AL	
			F4166: 00 000 NULL	ADD EBX + SI, AL	
			F4167: 00 000 NULL	ADD EBX + SI, AL	
			F4168: 00 000 NULL	ADD BH, BH	
			F4169: 00 000 NULL	DEC BP	
			F416A: 00 000 NULL	ADD AL, 0CFh	
			F416B: 00 000 NULL	ADD EBX + SI, AL	
			F416C: 00 000 NULL	ADD EBX + SI, AL	
			F416D: 00 000 NULL	ADD EBX + SI, AL	
			F416E: 00 000 NULL	ADD EBX + SI, AL	
			F416F: 00 000 NULL	ADD EBX + SI, AL	
			F4170: FF 255 RES	ADD BH, BH	
			F4171: FF 255 RES	DEC BP	
			F4172: CD 205 =	ADC BH, CL	
			F4173: 00 000 NULL	ADD EBX + SI, AL	
			F4174: CF 207 ±	ADD EBX + SI, AL	

Lab Task No 3(ii):

Solution:

Brief description (3-5 lines)

Arithmetic Expressions

An arithmetic expression is a mathematical statement that combines numbers and operators, like + (addition), - (subtraction), * (multiplication), or / (division), to perform calculations.

2. CX=8*9+7

The code

```

01 org 100h
02 MOV AX, 8
03 MOV AX, 9
04 ADD AX, 7
05 MOV CX, AX
06 MOV AH, 2
07 MOV DL, CH
08 ADD DL, '0'
09 INT 21h
10 MOV DL, CL
11 ADD DL, '0'
12 INT 21h
13 ret

```

The results (Screenshot)

registers		F400:0154	F400:0154
AX:	02 40	F4150: FF 255 RES	B108 D1
BX:	00 00	F4151: FF 255 RES	INT 020h
CX:	00 10	F4152: CD 205 =	INT
DX:	00 40	F4153: 20 032 SP0	ADD EBX + SI, AL
CS:	F400	F4154: CF 207 =	ADD EBX + SI, AL
IP:	0154	F4155: 00 000 NULL	ADD EBX + SI, AL
SS:	0700	F4156: 00 000 NULL	ADD EBX + SI, AL
SP:	FFFA	F4157: 00 000 NULL	ADD EBX + SI, AL
BP:	0000	F4158: 00 000 NULL	ADD EBX + SI, AL
SI:	0000	F4159: 00 000 NULL	ADD BH, BH
DI:	0000	F415A: 00 000 NULL	DEC BP
DS:	0700	F415B: 00 000 NULL	SBB CL, BH
ES:	0700	F415C: 00 000 NULL	ADD EBX + SI, AL
		F415D: 00 000 NULL	ADD EBX + SI, AL
		F415E: 00 000 NULL	ADD EBX + SI, AL
		F415F: 00 000 NULL	ADD EBX + SI, AL
		F4160: FF 255 RES	ADD BH, BH
		F4161: FF 255 RES	DEC BP
		F4162: CD 205 =	ADD BH, CL
		F4163: 1A 026 +	ADD EBX + SI, AL
		F4164: CF 207 =	ADD EBX + SI, AL
		F4165: 00 000 NULL	ADD EBX + SI, AL
		F4166: 00 000 NULL	ADD EBX + SI, AL
		F4167: 00 000 NULL	ADD EBX + SI, AL
		F4168: 00 000 NULL	ADD BH, BH
		F4169: 00 000 NULL	DEC BP
		F416A: 00 000 NULL	ADD AL, 0CFh
		F416B: 00 000 NULL	ADD EBX + SI, AL
		F416C: 00 000 NULL	ADD EBX + SI, AL
		F416D: 00 000 NULL	ADD EBX + SI, AL
		F416E: 00 000 NULL	ADD EBX + SI, AL
		F416F: 00 000 NULL	ADD EBX + SI, AL

Lab Task No 3(iii):

Solution:

Brief description (3-5 lines)

Arithmetic Expressions

An arithmetic expression is a mathematical statement that combines numbers and operators, like + (addition), - (subtraction), * (multiplication), or / (division), to perform calculations.

3. $CX = (-25 * 15) + 3 / 2 + 15 / 5$

The code

```

01 org 100h
02 MOV AX, -25
03 MOV AX, 15
04 MOV BX, AX
05 MOV AX, 3
06 MOV DX, 2
07 DIV DX
08 MOV DX, 15
09 MOV AX, 5
10 DIV DX
11 ADD AX, BX
12 ADD AX, DX
13 MOV CX, AX
14 ret

```

The results (Screenshot)

registers			F400:0174	F400:0174
AX	00	03	F4170: FF 255 RES	BIOS DI
BX	00	0F	F4171: FF 255 RES	INT 00h
CX	00	1F	F4172: CD 205 =	INT
DX	00	02	F4173: 00 000 NULL	ADD [BX + SI], AL
CS	F400		F4174: CF 207 ±	ADD [BX + SI], AL
IP	0174		F4175: 00 000 NULL	ADD [BX + SI], AL
SS	0700		F4176: 00 000 NULL	ADD [BX + SI], AL
SP	FFF8		F4177: 00 000 NULL	ADD [BX + SI], AL
BP	0000		F4178: 00 000 NULL	ADD [BX + SI], AL
SI	0000		F4179: 00 000 NULL	ADD BH, BH
DI	0000		F417A: 00 000 NULL	DEC BP
DS	0700		F417B: 00 000 NULL	ADD AL, 0CFh
ES	0700		F417C: 00 000 NULL	ADD [BX + SI], AL
			F417D: 00 000 NULL	ADD [BX + SI], AL
			F417E: 00 000 NULL	ADD [BX + SI], AL
			F417F: 00 000 NULL	ADD [BX + SI], AL
			F4180: FF 255 RES	ADD BH, BH
			F4181: FF 255 RES	DEC BP
			F4182: CD 205 =	ADC BH, CL
			F4183: 04 004 ♦	ADD [BX + SI], AL
			F4184: CF 207 ±	ADD [BX + SI], AL
			F4185: 00 000 NULL	ADD [BX + SI], AL
			F4186: 00 000 NULL	ADD [BX + SI], AL
			F4187: 00 000 NULL	ADD [BX + SI], AL
			F4188: 00 000 NULL	ADD BH, BH
			F4189: 00 000 NULL	DEC BP
			F418A: 00 000 NULL	ADC CL, BH